

Amanda Merkley

🏠 <https://a-merkley.github.io/> | www.linkedin.com/in/amandamerkley

Education

Carnegie Mellon University

PHD ELECTRICAL ENGINEERING

- Advisor: Dr. Pulkit Grover

Pittsburgh, PA

August 2021 - present

Colorado State University

BS COMPUTER ENGINEERING, BS MATHEMATICS

- GPA: 3.97 (*magna cum laude*)

Fort Collins, CO

May 2020

Research Experience

Carnegie Mellon University - Dept of Electrical & Computer Engineering

ADVISOR: DR. PULKIT GROVER

Pittsburgh, PA

August 2021 - Present

- *Variational autoencoder for latent interactions*. Design and implement a novel multi-layer generative model for neural signals.
- *Latent communication & causation in neural circuits*. Develop statistical model to identify latent communication and causal structure in neural systems.
- *Statistical analysis of noisy neural data*. Identify effects of Deep Brain Stimulation on high-dimensional neural activity.
- *High-order interactions in conflict processing*. Estimate O-information to show increased high-order redundancy in Stroop task.
- *Cognitive experiment design*. Design and run experiments on 20+ human epilepsy patients implanted with depth electrodes.

Sandia National Laboratories - Cognitive & Emerging Computing

ADVISOR: DR. WILLIAM CHAPMAN

Albuquerque, NM

June 2025-August 2025

- *Redundant & synergistic statistical information flow*. Extended information-theoretic redundancy-synergy index to directed information to compute high-order information flow between neural populations engaged in navigation.

Massachusetts Institute of Technology - Center for Brains, Minds, & Machines

ADVISOR: DR. NANCY KANWISHER

Boston, MA

June 2019-August 2019

- *Object permanence in humans*. Designed and ran experiments in object permanence, analyzed MEG signals of 5 subjects.

Publications

Y. Guo*, **A. Merkley***, S. Jaffee, A. C. Whiting, and P. Grover. "Identifying neural biomarkers of risk-taking from intracranial EEG recordings." IEEE EMBS Neural Engineering, 2025.

S. Jaffee, N. Gupta, D. Kramer, D. M. Kusk, J. Valeriano, **A. Merkley**, T. Kite, S. Arora, P. Grover, and A. C. Whiting. "Stereo-electroencephalography in the setting of a preexisting deep brain stimulation device: illustrative case." Journal of Neurosurgery: Case Lessons, 2025.

C. Goswami and **A. Merkley**. "Analytically deriving partial information decomposition for affine systems of stable and convolution-closed distributions." Advances in Neural Information Processing Systems, 2024.

A. Merkley and P. Grover. "Understanding neural population communication with latent channels." Allerton Conference on Communication, Control, and Computing, 2024.

A. Merkley, A. Y. Nam, Y. K. Hong, and P. Grover. "Message-relevant dimension reduction of neural populations." IEEE International Symposium on Information Theory, 2024.

Y. Li, S. Bhatnagar, **A. Merkley**, D. Weber, and S. Roy. "A predictor-corrector algorithm for fast polynomial chaos-based uncertainty quantification of multi-walled CNT interconnects." IEEE Trans on CPMT, 9(10), pp. 1963-1975, 2019.

*Equal contribution

S. Bhatnagar, **A. Merkley**, R. Berdine, Y. Li, and S. Roy, "Variability-aware performance assessment of multi-walled carbon nanotube interconnects using a predictor-corrector polynomial chaos scheme," in 2018 IEEE EDAPS, 2018.

Industry Experience

Western Digital Corporation

Longmont, CO

ASIC DEVELOPMENT ENGINEER

June 2020 - June 2021

- Verification of servo processor in enterprise hard drive ASIC using SystemVerilog and UVM

Presentations

October 2024. *Characterizing high-order interactions during conflict processing in patients with epilepsy*. Poster: Society for Neuroscience 2024, Chicago, IL.

September 2024. *Understanding neural population communication with latent channels*. Talk: Allerton 2024, Urbana-Champaign, IL.

July 2024. *Message-relevant dimension reduction of neural populations*. Talk: ISIT 2024, Athens, Greece.

June 2023. *Computing unique information for Poisson sources*. Poster: Decomposing Multivariate Info in Complex Systems, Dresden, Germany.

Awards & Fellowships

2025 **URA-Sandia Grad Student Summer Fellowship**, Oak Ridge Inst. for Science & Education
Waibel Presidential Fellowship, Carnegie Mellon University, School of Computer Science

2023 **Exemplary Qualifying Exam Performance**, Carnegie Mellon University, ECE Department

2022 **NSF Graduate Research Fellowship Program**, National Science Foundation
Fritsch Family Fellowship, Carnegie Mellon University, ECE Department

2021 **NIH T-32 Neural Interfacing Traineeship**, National Institutes of Health

2019 **Astronaut Scholarship**, Astronaut Scholars Foundation
Goldwater Scholarship, Goldwater Foundation
Richard E. Merwin Student Scholarship, IEEE Computer Society

Service & Teaching

Summer 2024 **Summer Internship Mentor**, Mentored 2 undergraduate students

July 2024 **ISIT 2024 Neuroscience + Information Theory Workshop**, Workshop co-organizer

Spring 2024 **Teaching Assistant**, Information Theory Measures (18-753)

2021-2023 **ECE Graduate Organization**, Carnegie Mellon University, Treasurer

2017-2019 **IEEE Student Chapter**, Colorado State University, President